SET 2: THE AEGAEON RIDGE LUXURY HABITATION & RESEARCH SPIRES

DOCUMENT 2A: SUBORDINATED MEZZANINE CREDIT AGREEMENT

SUBORDINATED MEZZANINE LOAN AGREEMENT

Dated as of the Closing Date (as defined herein)

among

AEGAEON HABITATION PARTNERS LLC,
as Borrower

and

OFF-WORLD REAL ESTATE OPPORTUNITIES FUND LP,
as Lender

and

LUNAR-MARTIAN TRUST COMPANY,
as Fiscal Agent

ARTICLE I: DEFINITIONS AND ACCOUNTING TERMS

Section 1.01. Defined Terms. As used in this Agreement, the following terms have the meanings specified below:

- **"Closing Date"** means the date on which this Agreement is executed by all participating parties, as designated next to each signature in the signature blocks below (such execution date, the **"Signature Date"**).

- **"Decompression Mitigation Protocol"** means the deployment sequence of rapid aerosol-based self-healing foam sealant into structural stress points of the habitation spires.

- **"He-3 Supply Index"** means the monthly unit price of Helium-3 isotope canisters imported from lunar orbital suppliers, as used to power the local secondary backup generators.

- **"Pressurized Volume Unit"** or **"PVU"** means one million cubic meters of gas mix at standard Earth surface density, containing not less than 20.9% oxygen (O_2), 78.1% nitrogen (N_2), and less than 1.0% carbon dioxide (CO_2).

- **"Spires"** means the three interconnected structural vertical habitation towers (Spire Alpha, Spire Beta, and Spire Gamma) suspended along the gravity-shear cliffs of Aegaeon Ridge, Titan.

ARTICLE II: THE MEZZANINE FACILITY

Section 2.01. The Mezzanine Loan. Subject to the terms and conditions set forth herein, the Lender agrees to make a single advance Subordinated Mezzanine Loan to the Borrower on the **Closing Date** in a principal amount equal to **1,200,000,000 Sol Credits (SC)**.

Section 2.02. Interest and Subordination.

1. **Interest Rate:** The Loan shall bear interest at a fixed rate of **13.50% per annum**, compounded quarterly.

2. **Payment-in-Kind (PIK) Election:** If the Borrower's Cash-on-Hand is less than SC 50.0 million due to unexpected micro-meteoroid armor repairs, the Borrower may elect to pay up to 5.0% of the interest due in PIK notes, in which case such interest will be capitalized into the principal outstanding balance of the Loan.

3. **Subordination:** The payment of principal and interest on the Mezzanine Loan is fully subordinated in right of payment to the Senior Credit Agreement held by the Jovian-Saturnine Colonial Development Bank.

ARTICLE III: ENVIRONMENTAL AND TECHNICAL REPRESENTATIONS

The Borrower represents and warrants to the Lender that:

Section 3.01. Life-Support Reliability. The Spires' centralized life-support closed-loop recycling systems are rated to support up to 12,000 permanent human occupants at an average metabolic load of 2,500 kcal per day, with active oxygen and water reclamation efficiency of no less than 99.8%.

Section 3.02. Radiation Protection Index. The electro-magnetic deflective shielding envelope surrounding the Spires is fully functional and capable of deflecting high-energy cosmic rays and Saturnian magnetospheric energetic particles down to safe occupational limits (< 20 mSv/Earth-year equivalent).

ARTICLE IV: COVENANTS

Section 4.01. Financial Covenants.

1. **Debt Service Coverage Ratio (DSCR):** The Borrower shall maintain a DSCR of not less than **1.15x** at the end of each fiscal quarter, calculated as Net Operating Cash Flow divided by the sum of Senior Debt Service and Mezzanine Debt Service.

2. **Occupancy Covenant:** The Borrower shall maintain a weighted-average physical occupancy rate of not less than **72.5%** across the Research Spires (measured by leased **PVUs**) and not less than **60.0%** across the Elite Residential Penthouses.

ARTICLE V: COVENANTS ON DEFAULT

Section 5.01. Immediate Acceleration Event. Upon the occurrence of any "Life-Support De-certification" issued by the Sol-System Health and Human Safety Directorate, or upon any physical decompression event resulting in the complete evacuation of any individual Spire, the outstanding principal and all accrued interest under this Agreement shall immediately become due and payable.

AEGAEON HABITATION PARTNERS LLC

By: _____
Name: Julian Kase
Title: Managing Director

Date: **June 26, 2146**

OFF-WORLD REAL ESTATE OPPORTUNITIES FUND LP

By: _____
Name: Michael Thorne
Title: Portfolio Lead (Saturn Region)

Date of Execution (the **Signature Date**): **June 26, 2146**